

MUHAMMAD MUTAHHAR BAIG

Embedded Systems Engineer | Embedded AI & IoT

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SUMMARY

Embedded systems engineer with hands-on experience building firmware and edge-AI products, from wearables and medical devices to IoT controllers and wireless audio systems. Comfortable across ESP32/STM32/Zephyr firmware, BLE/Wi-Fi/MQTT connectivity, and deploying ML models (TensorFlow Lite, ONNX, RKNN) on constrained hardware. Electronics Engineering graduate, UET Taxila.

WORK EXPERIENCE

Embedded Systems & AI Engineer | Khayal IncoTech

Dec 2025 – Present

- Designed and shipped firmware for several ESP32-S3 audio and IoT products, covering real-time I2S audio capture/playback, SD-card storage, BLE/Wi-Fi provisioning, and OTA update systems with automatic rollback.
- Built a Bluetooth-based wearable monitoring device on a Nordic nRF52 SoC (Zephyr RTOS), including a real-time audio signal-processing pipeline and custom BLE services with over-the-air updates.
- Developed an IoT controller with dual BLE and MQTT connectivity, automatic network fallback, and a custom communication protocol for local and remote device control.
- Built a wrist-worn heart-rate monitor on ESP32-S3, implementing real-time PPG signal processing and a live monitoring dashboard.
- Led acoustic R&D for a digital stethoscope, evaluating microphone hardware and building signal-processing and machine-learning pipelines for heart-sound analysis.
- Deployed and benchmarked speech-to-text pipelines across 8+ APIs and open-source models, with production-ready REST API integration.
- Engineered power-optimization strategies (dynamic clock scaling, radio sleep states, deep-sleep logic) to extend battery life on low-power devices.

Embedded Systems Engineer | MachaDev Engineering, Air University Islamabad

Jul 2025 – Nov2025

- Designed and developed embedded systems and electronic hardware across multiple microcontroller/microprocessor platforms (ESP32, STM32, CH32, Raspberry Pi, Rockchip RK3566/3576/3588).
- Trained, optimized, and deployed ML/DL models to edge devices, including quantization, pruning, and TensorFlow Lite/ONNX/RKNN conversion for constrained hardware.
- Built an on-device AI system for real-time image classification with NPU-accelerated inference, plus a multi-modal audio pipeline combining sound classification, speech-to-text, and text-to-speech.
- Developed real-time computer-vision pipelines for object and impact detection, combining classical CV techniques with a custom-trained deep learning model.
- Developed real-time device software using C/C++, Python, and MATLAB.

AI Intern | NICAT (National Incubation Centre for Agri-Tech), NASTP

Aug 2024 – Feb2025

- Developed and optimized an AI-powered weed detection system for precision agriculture, reducing herbicide usage through targeted application.
- Implemented model optimization techniques for real-time deployment on edge devices under field constraints.

FINAL YEAR PROJECT

Design & Development of Agri-Rover for Crop Management

- Designed the mechanical and electronic architecture of an AI-powered agricultural rover for crop health monitoring, with multi-sensor fusion and >90% detection accuracy.
- Built fully autonomous GPS navigation with three operating modes; selected for the All-Punjab Expo Competition 2025.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Verilog, Java

AI/ML: PyTorch, TensorFlow, TensorFlow Lite, OpenCV, MediaPipe, Model Training & Deployment, Quantization, Pruning, ONNX, RKNN

Embedded Systems: ESP32, STM32, CH32, Arduino, Raspberry Pi, Rockchip (RK3566/3576/3588), Nordic nRF52, FPGA, Pixhawk, J-Link

Connectivity: BLE, Wi-Fi, MQTT, I2S, I2C, SPI, UART
Dev Tools: ESP-IDF, Zephyr/nRF Connect SDK, MounRiver Studio, Arduino IDE, MATLAB

EDUCATION

Bachelor of Science in Electronics Engineering UET, Taxila	Nov 2021 – Jun 2025
Masters in Artificial Intelligence NUST	Sep 2026 – Present

ACHIEVEMENTS & MEMBERSHIPS

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- IEEE Student Member Consumer Technology Society, Engineering in Medicine and Biology Society
 - Winner, Open House 2025, Electronics Engineering Department
 - Final Year Project selected for All-Punjab Expo Competition 2025; 1 conference paper submitted
 - Hafiz-e-Quran